Project in Statistics: High-Dimensional Inference for Ornstein-Uhlenbeck Processes

Frederik Mølkjær Andersen (vrs751)
Supervisor: Dmytro Marushkevych

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1 Introduction

In this project we are concerned with high-dimensional sparse inference from continuously observed stochastic processes of a particular form, namely those who solve the linear stochastic differental equation for a sparse matrix A,

$$dX_t = -AX_t\,dt + dW_t$$

called an Ornstein-Uhlenbeck diffusion processes. The main part of the project is a review of estimators which are more or less useful for estimating A given a continuous observation of the process.

We start by developing the necessary theory for diffusion processes and turn to classical maximum likelihood estimation, which we will demonstrate is not suitable for sparse and high-dimensional problems due to the curse of dimensionality. Thus we consider more restrictive estimators, which gains power in estimation at the cost of assuming sparsity in the parameter. These include the LASSO, some variations on the LASSO and the Dantzig estimator.

We will present these estimators and essential theoretical results and considerations, but will focus on simulation studies to see how the estimators behave in practice. A large part of the project has been implementation of the different estimators, which are likewise described along the way. A companion Quarto-notebook follows along containing the main functions for simulation and estimation.

1.1 Notation and underlying assumptions

We will briefly outline some notation. A vector $x \in \mathbb{R}^k$ is a column vector and its transpose $x^\intercal$ is a row vector. The identity matrix in $\mathbb{R}^d$ are denoted I_d. When considering the ℓ_p-norm for $p \geq 1$ of a matrix $A \in \mathbb{R}^{d_1 \times d_2}$ we mean the entrywise ℓ_p-norm,

$$\|A\|_p = \left(\sum_{i \leq d1, j \leq d_2} |A_{ij}|^p \right)^{1/p}$$

and note that $\|\cdot\|_2 = \|\cdot\|_F$ where $\|\cdot\|_F$ is the Frobenius norm. We further let $\|\cdot\|_\infty = \lim_{p\to\infty} \|\cdot\|_p = \max\{|A_{ij}| : i \leq d_1, j \leq d_2\}$ and $\|\cdot\|_0 =$

$\sum_{i \le d_1, j \le d_2} 1_{\{A_{ij} \neq 0\}}$. Note however that $\|\cdot\|_0$ is not actually a norm, likewise we can define functions $\|\cdot\|_p$ just as above for $0 < p < 1$ which are not actually norms, but may still be useful.

Furthermore $\circ$ denotes Hadamard-like operations on matrices, e.g. entrywise multiplication and exponentiation, $\otimes$ denotes the Kronecker product, and vec denotes vectorization of a matrix by stacking its columns.

We use $X \sim P$ to denote that X has distribution P and $x \overset{s}{\sim} P$ to denote that x is a sample from P. $\overset{P}{\to}$, $\overset{a.s.}{\to}$ and $\overset{\mathcal{D}}{\to}$ denotes convergence in probability, almost sure convergence and weak convergence (convergence in distribution) respectively. $N(\mu, \Sigma)$ is a d-dimensional Gaussian distribution with mean-vector $\mu \in \mathbb{R}^d$ and covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$.

We shall not go into too many measure-theoretic details in this report, but for completeness we will assume that all stochastic processes defined in this report are silently assumeed to be adapted processes defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \ge 0}, \mathbb{P})$ which may be completed, i.e. for every set $A \subset B \in \mathcal{F}$ where $\mathbb{P}(B) = 0$ is itself in the σ-algebra $A \in \mathcal{F}$. When considering estimators we leave out the measure-theoretical details too, but essentially these are measurable mappings from the sample space (i.e. taking as a argument the observation) into the parameter space (giving a estimate from the observation).

Whenever we mention an observed path of a process we theoretically mean a continuous observation, which of course, at least on a computer, is not possible. Hence i practice we work with finely discretized observations of the process.

We will throughout the report define a number of assumptions (H1-3) as we need them. After they are defined we will silently assume them unless otherwise stated. The assumptions are inspired by the ones in [1, 2].

2 Brownian motion

2.1 Definition and elementary properties

Definition 2.1 *Let $(W_t)_{t \ge 0}$ be a continuous time stochastic process with values in $\mathbb{R}$. We call $(W_t)_{t \ge 0}$ a (standard) Brownian Motion or Wiener Process if*

1. *$W_0 = 0$ almost surely,*
2. *$W_t - W_s \sim N(0, t - s)$ for all $t > s \ge 0$,*
3. *The process has independent increments, i.e. for $0 < t_1 < \cdots < t_n$, $W_{t_1}, W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}}$ are mutually independent,*
4. *Sample paths $t \mapsto W_t$ are almost surely continuous.*

We will also call the process $(W_t)_{t \in [0,T]}$ for some $T > 0$ a Brownian motion.

The following moment results follows easily from the definition,

Lemma 2.2 *Let $(W_t)_{t \ge 0}$ be a Brownian motion. For all $t, s \ge 0$*

$$\mathbb{E}W_t = 0, \mathbb{E}W_t^2 = t \text{ and } \operatorname{Cov}(W_t, W_s) = \mathbb{E}[W_t W_s] = \min\{t, s\}.$$

Whether or not a process as in Definition 2.1 actually exists is not clear. The problem lies in whether the restrictions we have made on the finite-dimensional distributions of the process allows for the sample paths to be continuous. It is however possible to construct a Brownian motion for example through a uniform limit of combinations of (cleverly

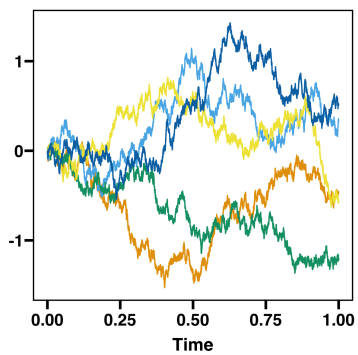

Figure 1: 5 sample paths of a Brownian motion.

chosen) continuous functions, guaranteeing that the resulting process exists and has continuous paths.

Theorem 2.3 *There exists a Brownian motion (which have almost surely continuous sample paths). ([3], e.g. Section 2.3).*

The sample paths of a Brownian motion have a number of interesting properties, but for the purpose of this project we will only mention that the paths are almost surely non-differentiable,

Theorem 2.4 *A Brownian motion is almost surely nowhere differentiable. ([3], Theorem 9.18).*

Remark 2.5 Our purpose for studying Brownian motions are to work with stochastic differential equations like $dX_t = b(t, X_t)\,dt + \sigma(t, X_t)\,dW_t$. With Theorem 2.4 in mind, this does not look promising as dW_t does not exists in a traditional derivate sense. However, as we will remark later (Remark 3.2), the SDE above is nothing but shorthand notation for a combination (stochastic) integrals which we are able to define. Meanwhile, we will consider $(dW_t)_{t\geq 0}$ as being a generalized stochastic process called the *white noise process*, where generalized means that the process can be constructed as a probability measure on a larger space of tempered distributions on $[0, \infty)$ than $\mathbb{R}^{[0,\infty)}$ [4].

One could further consider a Brownian motion with drift,

Definition 2.6 *Let $(W_t)_{t\geq 0}$ be a Brownian motion, $\mu \in \mathbb{R}$ and $\sigma > 0$, a Brownian motion with drift is then a process $(W_t^D)_{t\geq 0}$ with*

$$W_t^D = \mu t + \sigma^2 W_t \quad \textit{for all } t \geq 0.$$

Moment results follows immediately from the definition,

Corollary 2.7 *Let $(W_t^D)_{t\geq 0}$ be a Brownian motion with drift,*

$$\mathbb{E}W_t^D = \mu t, \quad \mathrm{Cov}(W_t^D, W_s^D) = \sigma^2 \min\{t, s\}. \tag{1}$$

One reason why Brownian motions play a central role in probability and statistics is that they arise as the (weak) limit in a functional version of the central limit theorem. We are not particularly interested in this limit theorem as such, but will use it for numerical simulations and thus present it (although in a weaker than necessary version).

Consider a sequence $\xi_1, \xi_2, \ldots$ of i.i.d. random variables with $\mathbb{E}\xi_1 = 0$ and $\sigma^2 = \mathrm{Var}\,\xi_1 < \infty$ and let these be the increments of a random walk $S_n = \sum_{i=1}^n \xi_i$ with $S_0 = 0$. We can now define a continuous time stochastic process $(Y_t)_{t\geq 0}$ as the linear interpolation of the random walk,

$$Y_t = S_{[\![t]\!]} + (t - [\![t]\!])\xi_{[\![t]\!]+1}$$

for $t \geq 0$, where $[\![t]\!]$ is the greatest integer less than or equal to t. We will now define the sequence of processes $\{(X_t^{(n)})_{t\geq 0}\}_{n\in\mathbb{N}}$, as rescaled versions of $(Y_t)_{t\geq 0}$,

$$X_t^{(n)} = \frac{1}{\sqrt{\sigma^2 n}} Y_{nt} \tag{2}$$

for $t \geq 0$ and $n \in \mathbb{N}$.

Theorem 2.8 *Let $\{(X_t^{(n)})_{t\geq 0}\}_{n\in\mathbb{N}}$ be given by (2) and let $0 \leq t_1 < t_2 < \cdots <$*

$t_d < \infty$, *then for all* $t, s \geq 0$

$$(X_{t_1}^{(n)}, X_{t_2}^{(n)}, \ldots, X_{t_d}^{(n)}) \xrightarrow{\mathcal{D}} (W_{t_1}, W_{t_2}, \ldots, W_{t_d})$$

as $n \to \infty$, *where* $(W_t)_{t\geq 0}$ *is a Brownian motion. ([3], Theorem 4.17)*

The full theorem, called *Donsker's invariance principle* or *the functional central limit theorem* further states that the whole sequence of rescaled random walks converge in distribution to a Brownian motion $\{(X_t^{(n)})_{t\geq 0}\} \xrightarrow{\mathcal{D}} (W_t)_{t\geq 0}$. For more details see [3].

2.2 d-dimensional Brownian motion

Definition 2.9 *Let* $d \in \mathbb{N}$ *and* $(W_t)_{t\geq 0}$ *a continuous time stochastic process with values in* $\mathbb{R}^d$. *We call* $(W_t)_{t\geq 0}$ *a d-dimensional Brownian motion if*

1. $W_0 = 0$ *almost surely,*
2. $W_t - W_s \sim N(0, (t-s)I_d)$ *for all* $t > s \geq 0$,
3. *The process has independent increments, i.e. for* $0 < t_1 < \cdots < t_n$, $W_{t_1}, W_{t_2} - W_{t_1}, \ldots, W_{t_n} - W_{t_{n-1}}$ *are mutually independent,*
4. *Sample paths* $t \mapsto W_t$ *are almost surely continuous.*

We will also call the process $(W_t)_{t\in[0,T]}$ *for some* $T > 0$ *a d-dimensional Brownian motion.*

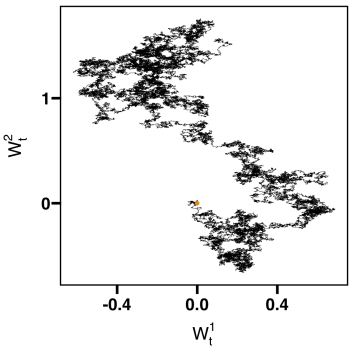

Figure 2: Sample path from a 2-dimensional Brownian motion.

Remark 2.10 Definition 2.1 and Definition 2.9 obviously looks very much alike, in particular notice that since $W_t - W_s \sim N(0, (t-s)I_d)$ where the covariance matrix is diagonal, a d-dimensional Brownian motion can be constructed as d independent 1-dimensional Brownian motions, this also ensures that the properties discussed up until now are easily extended to the d-dimensional processes [3].

The two dimensional Brownian motion was historically introduced as a model for the movement of pollen suspended in water in which the pollen are moving around in an irregular pattern similar to Figure 2 because of its constant collision with the the water molecules. It has many applications in physics, biology and of course as one of the most fundamental stochastic processes in mathematics. We will use it to study more intricate stochastic processes namely those given by a certain class of stochastic differential equations.

3 Ornstein-Uhlenbeck processes

The main goal of this project is to draw inference from a particular class of stochastic differential equations, namely those who define an Ornstein-Uhlenbeck process. We start by considering the SDE in dimension $d = 1$.

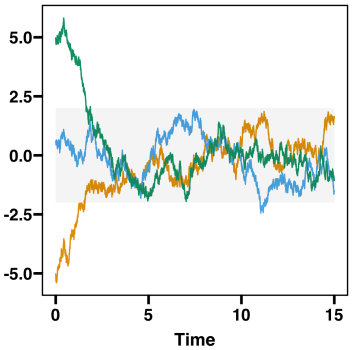

Figure 3: Sample trajectories of an OU process with positive drift parameter ($\theta = 0.5, \sigma = 1$), two started away from concentration of the stationary distribution (orange and green, started in $x_0 = -5, 5$ respectively) and one started in the concentration (blue). The shaded area shows a 95% prediction interval for the stationary distribution.

3.1 1-dimensional Ornstein-Uhlenbeck process

Definition 3.1 *Let* $(X_t)_{t\geq 0}$ *be an 1-dimensional real stochastic process (on the filtered space* $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$*), we call* $(X)_{t\geq 0}$ *an Ornstein-Uhlenbeck process if for all* $t \geq 0$, $X_t \in \mathbb{R}$ *is the solution to*

$$dX_t = -\theta X_t \, dt + \sigma \, dW_t \tag{3}$$

given starting point $X_0 = x_0 \in \mathbb{R}$, *where* $\theta \in \mathbb{R}$, $\sigma > 0$ *and* $(W_t)_{t\geq 0}$ *is a Brownian motion.*

Remark 3.2 The formulation $dX_t = -\theta X_t\, dt + \sigma\, dW_t$ does not make sense at first sight since we have seen in Theorem 2.4 that dW_t/dt does not exists almost surely for all $t \in \mathbb{R}$, but (3) is nothing more than short-hand notation for

$$X_t = X_0 - \theta \int_0^t X_t\, dt + \sigma \int_0^t dW_t \tag{4}$$

where the last integral is an Itô integral (for more details see e.g. [4]).

The process is mean-reverting when $\theta > 0$, which (informally) is due to the drift term being dependent on the current value of the process, such that when the process is below its asymptotic mean (in our definition the asymptotic mean is 0) the drift will be positive, and when above its asymptotic mean the drift will be negative. Now consider a Brownian motion with drift, we clearly see the contrast with the OU process as the drift in a Brownian motion is constant which in particular means that the process is not mean-reverting.

Not only is the process mean-reverting when $\theta > 0$, it is also ergodic and has stationary distribution $N(0, \sigma^2/2\theta)$. This will be important for the estimation procedures which are to follow but the effects of whether $\theta > 0$ is also quite visible in simulated trajectories. From Figures 3 and 4 we see that the trajectories from an OU process with positive drift quickly reverts from the starting points (chosen away from the concentration of the stationary distribution) towards the mean, while the trajectories from the process with negative drift are not driven towards the mean indicating (lack of) mean reversion of the process. The figures also support that there exists a stationary distribution for the process with $\theta > 0$, while no such (at least with finite variance) exists for the $\theta < 0$. Furthermore we see from Figure 5 that the process with $\theta > 0$ actually satisfies the ergodicity condition while the process with $\theta < 0$ does not.

The Ornstein-Uhlenbeck process was historically a model of the velocity of a Brownian particle under friction (acceleration of $-\theta$ times the velocity X_t). But the general nature of the model as a mean-reverting process has made it one of the most used processes in areas such as financial modeling and biology.

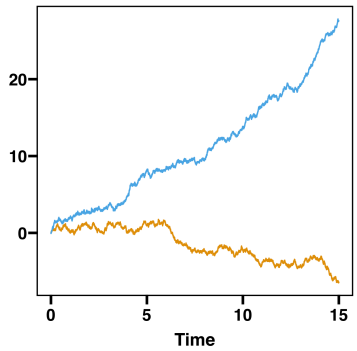

Figure 4: Sample trajectories from OU process with non-positive drift parameter ($\theta = -0.1, \sigma = 1$).

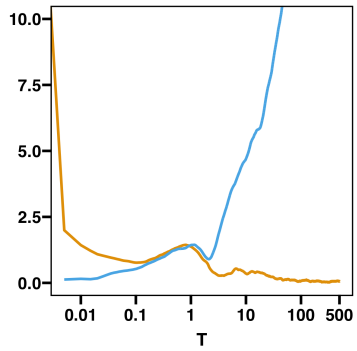

Figure 5: $\left\| \frac{1}{T} \int_0^T X_t dt \right\|_F$ for sample trajectories of a process under (H1) (orange) and one not under (H1) (blue) 5-dimensional OU process. (Time axis is on $\log_{10}$-scale)

3.2 d-dimensional Ornstein-Uhlenbeck process

The Ornstein-Uhlenbeck process can naturally be extended to take values in $\mathbb{R}^d$ for any $d \in \mathbb{N}$,

Definition 3.3 *Let $(X_t)_{t\geq 0}$ be a d-dimensional real stochastic process, we call $(X_t)_{t\geq 0}$ a d-dimensional Ornstein Uhlenbeck process if for all $t \geq 0$, $X_t \in \mathbb{R}^d$ is the solution to*

$$dX_t = -AX_t\, dt + dW_t \tag{5}$$

given starting point $X_0 = x_0 \in \mathbb{R}^d$, with $A \in \mathbb{R}^{d\times d}$ and $W = (W_t)_{t\geq 0}$ being a d-dimensional Brownian motion.

Remark 3.4 More generally one could have the diffusion part of the SDE be a linear transformation of the Brownian Motion, i.e. we could have $dX_t = -AX_t\, dt + S\, dW_t$ for some invertible $S \in \mathbb{R}^{d\times d}$, analogously to $\sigma > 0$ in Definition 3.1. For reasons which will become clear in Remark 4.1 we will only consider the OU process as in Definition 3.3 where $S = I_d$.

Analogous to the 1-dimensional case, results on mean-reversion, ergodicity and existence of a stationary distribution depends on the drift

parameter which is now a matrix. A natural and analogous condition would be to assume that A has strictly positive eigenvalues, and in fact this will ensure those properties. More generally we consider the assumption (H1) which we will assume throughout the report unless otherwise stated,

(H1) A is diagonisable with eigenvalues $\alpha_1, \ldots, \alpha_d$ satisfying $\Re(\alpha_i) > 0$ for each $i = 1, \ldots, d$. (Note that this boils down to $\theta > 0$ for $d = 1$.)

Theorem 3.5 *Under **(H1)** X is ergodic, mean-reverting and have stationary distribution $N(0, C_\infty)$ where*

$$C_\infty = \int_0^\infty \exp(-sA)\exp(-sA^\top)\, ds. \tag{6}$$

(Note that this result for 1-dimensional and d-dimensional processes coincide.)

To see the effect of assumption (H1) on sample paths see Figure 6 and Figure 7, here we see that the process under (H1) returns to its stationary distribution while the one not satisfying the assumption drifts off infinitely and does not attain a stationary state.

Like with the 1-dimensional Ornstein-Uhlenbeck process, the multidimensional variant also has applications in areas as financial modeling and biology. One interesting application considers A as being an adjacency matrix in a network of many linearly interacting nodes. Such networks will in practice seldom be fully connected but rather sparse, therefore it will be interesting to consider estimation of high-dimensional sparse drift matrices from an observed high-dimensional Ornstein-Uhlenbeck process.

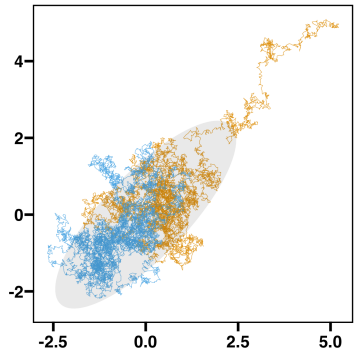

Figure 6: Two sample paths from an ergodic 2-dimensional OU process, the blue one is started in the concentration stationary distribution, while the orange one is started outside. The grey ellipse is a 95% prediction area for the stationary distribution.

3.3 Simulation

As the goal of this project is to compare the performance of a number of estimators of the drift matrix A in an observed Ornstein-Uhlenbeck process, we will consider simulation studies where A is known and the observed process is simulated. Thus we will consider the question of actually simulating from an Ornstein-Uhlenbeck process (a procedure which is essentially valid for all SDE).

3.3.1 Euler-Maruyama Scheme

Given a matrix $A \in \mathbb{R}^{d\times d}$ let us approximate a sample path from $(X_t)_{0\le t\le T}$ solving $dX_t = -AX_t\, dt + dW_t$ over $[0, T]$ with starting point $X_0 = x_0$. A simple solution, given a discretization of $[0, T]$, $0 = \tau_0 < \tau_1 < \cdots < \tau_n < \cdots < \tau_N = T$, is to simulate a (continuous time) process $(Y_t)_{0\le t\le T}$ satisfying the Euler-Maruyama scheme,

$$Y_{\tau_{n+1}} = Y_{\tau_n} - AY_{\tau_n}(\tau_{n+1} - \tau_n) + (W_{\tau_{n+1}} - W_{\tau_n}) \tag{7}$$

for $n = 1, \ldots, N-1$ and $Y_{\tau_0} = x_0$. We will only consider equidistant disretizations, i.e $\Delta\tau_n \equiv \delta = T/N$ where N is reasonably large.

Hence simulating the discrete time embedded process $(Y_{\tau_n})_{n=0,1,\ldots,N}$ is a question of iteratively computing the deterministic increments $-AY_{\tau_n}\delta$ and sampling the increment of a Brownian motion which we have seen

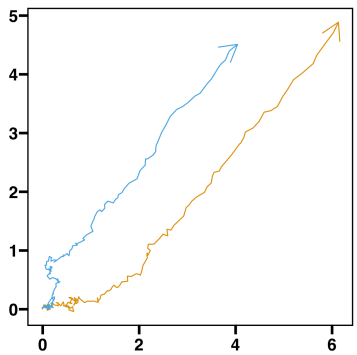

Figure 7: Two sample paths from an non-ergodic 2-dimensional OU process both started in $(0, 0)$.

in Definition 2.1 are Gaussian, $W_{\tau_{n+1}} - W_{\tau_n} \sim N(0, \tau_{n+1} - \tau_n) = N(0, \delta)$ [5].

If we want to extend to the full continuous time approximating process $(Y_t)_{0 \le t \le T}$ we could consider the piecewise constant process $Y_t = Y_{\tau_{n_t}}$ where $n_t = \max\{n : \tau_n \le t\}$ or we could linearly interpolate,

$$Y_t = Y_{\tau_{n_t}} + \frac{t - \tau_{n_t}}{\tau_{n_t+1} + \tau_{n_t}} (Y_{\tau_{n_t+1}} - Y_{\tau_{n_t}})$$

which ensures that $(Y_t)_{0 \le t \le T}$ is continuous. The linear interpolation is used in the plots of sample trajectories in this report. For more details on simulating paths from SDEs see e.g. [5].

3.3.2 Generation of drift matrix

We are interested in estimation from observed ergodic Ornstein-Uhlenbeck processes and we should thus simulate the process using a drift matrix satisfying (H1). As also motivated we will consider drift matrices with sparse entries, i.e. a high number of the entries are zero. Let the density of non-zero elements of a matrix $A \in \mathbb{R}^{d \times d}$ be $\rho(A) = d^{-2} \sum_{i,j \le d} 1_{A_{ij} \neq 0} \in [0, 1]$, then sparsity means that $\rho(A) << 1$. If we do not want to impose any further structure, such as symmetry, on the matrix, no general and non-trivial procedure of generating such matrices exists to the knowledge of the author.

A simple method is implemented. Starting with a diagonal matrix with positive entries only (hence satisfying (H1)), we add random entries at random places in the matrix and check whether or not this proposed matrix satisfies (H1). When the desired density of non-sparse entries are reached the procedure is stopped. We have to supply the method with the dimension of the matrix, desired density ρ and maximal absolute element size a_{max}. The random entries are then samples from $\text{Unif}(-a_{max}, a_{max})$. See Algorithm 1. The method is not particularly efficient, but for the purpose of the simulation studies in this project it is reasonable.

Algorithm 1: Algorithm for generating random and sparse matrix with non-zero element density $\approx \rho^*$ satisfying (H1).

Initialize: A, A^{proposal} $d \times d$ matrix;
for *i=1,...,d* **do**
  $A_{ii} \overset{s}{\sim} \text{Unif}(-a_{\max}, a_{\max})$;
while $\rho(A) \le \rho^*$ **do**
  $(i, j) \overset{s}{\sim} \text{Unif}(\{1, \dots, d\} \times \{1, \dots, d\})$;
  $A^{\text{proposal}} \leftarrow A$;
  $A^{\text{proposal}}_{ij} \overset{s}{\sim} \text{Unif}(-a_{\max}, a_{\max})$;
  if $\Re(\alpha_i(A^{\text{proposal}})) > 0 \, \forall i = 1, \dots, d$ **then**
    $A \leftarrow A^{\text{proposal}}$

3.3.3 Drawing from stationary distribution

For better performance of the following estimators we will in (H2) assume that the observed Ornstein-Uhlenbeck process are started in its stationary distribution, i.e. $X_0 = x_0 \overset{s}{\sim} N(0, C_\infty)$ given that A satisfies (H1). That leaves us with the problem of sampling from this distribution. For this,

two approaches are considered, either we can compute C_∞ from Theorem 3.5 or we can simulate a sample path from the process until stationarity.

Both methods come with their advantages and disadvantages. Sampling from the stationary distribution by simulating the process for a long time is reasonably efficient but ensuring that the process has actually attained stationarity is not straightforward. If C_∞ were known explicitly sampling from $N(0, C_\infty)$ could easily be done, but the integral (6) have to be discretized and it is an improper integral from 0 to ∞. Thus computing C_∞ is not efficient for high d.

4 Maximum likelihood estimation

Our first step towards estimating the drift matrix in a high-dimensional Ornstein-Uhlenbeck process is to once again consider the 1-dimensional process. So in the following exposition let $(X_t)_{t\geq 0}$ be a solution to $dX_t = -\theta X_t\, dt + dW_t$ for $\theta \in \mathbb{R}$ and $(W_t)_{t\geq 0}$ being a 1-dimensional Brownian motion. Suppose that we have observed the process $(X_t)_{t\in[0,T]}$ continuously. For simplicity of the following results, and since it is in many practical cases a natural assumption, we will assume for the remainder of the report that the observed process has already attained its stationary distribution,

(H2) $X_0 \sim N(0, C_\infty)$.

Under (H2) we in particular have that $X_t \sim N(0, C_\infty)$ for all $t \geq 0$.

Remark 4.1 We could also consider a rescaled diffusion part, i.e. letting $\sigma \neq 1$ in Definition 3.1, but using that for diffusion processes such as the OU process

$$\sum_{i=1}^{n} \left(X_{iT2^{-n}} - X_{(i-1)T2^{-n}}\right)^2 \to \sigma^2 T \quad a.s. \quad \text{for } n \to \infty$$

(see [6], p. 212) we could from our continuous observation of $(X_t)_{t\geq 0}$ (a.s.) compute (not estimate) σ. We will therefore not consider estimation of the diffusion part of our SDEs.

Now the maximum likelihood estimator has the usual definition,

Definition 4.2 *Let P_θ^T, P_W^T be the probability measures generated by respectively $(X_t)_{0\leq t\leq T}$ and $(W_t)_{0\leq t\leq T}$, (we have that $P_\theta^T << P_W^T$ [6]). The maximum likelihood estimator (MLE) $\hat{\theta}_T$ is such that the Radon-Nikodym derivative at $\hat{\theta}_T$ satisfies,*

$$\frac{dP_{\hat{\theta}}^T}{dP_W^T} = \sup_{\theta\in\Theta} \frac{dP_\theta^T}{dP_W^T}.$$

The MLE exists almost surely under (H1) and we can further write down the negative log-likelihood using Girsanov's Theorem [7],

$$\ell_T(\theta) = -\log\left(\frac{dP_\theta^T}{dP_W^T}\right) = \int_0^T \theta X_t\, dX_t + \frac{1}{2}\int_0^T (\theta X_t)^2 dt$$

Finding the derivative $\ell_T'(\theta)$ and solving the score equation $\ell_T'(\theta) = 0$

then gives us the MLE,

$$\hat{\theta}_T = -\frac{\int_0^T X_t \, dX_t}{\int_0^T X_t^2 \, dt} = \theta - \frac{\int_0^T X_t \, dW_t}{\int_0^T X_t^2 \, dt}$$

where the last equality is useful for asymptotic analysis of $\hat{\theta}_T$, while the first is what is used for actual estimation. A number of asymptotic results holds for the MLE if (H1) is satisfied (that is $\theta > 0$), in particular $\hat{\theta}_T$ is consistent $\hat{\theta}_T \xrightarrow{a.s.} \theta$ and asymptotically normal $\sqrt{T}(\hat{\theta}_T - \theta) \xrightarrow{\mathcal{D}} N(0, 2\theta)$.

4.0.1 A simulation study

To assert these asymptotic properties of $\hat{\theta}_T$ we conduct a simulation study. Choosing $\theta = 4$ we sample 100 independent discretized paths $(x_t)^i_{0 \le t \le 1000}$ from $(X_t)_{0 \le t \le 1000}$ where $i = 1, \ldots, 100$ with a step size of $\Delta t = 0.001$. For each path we make 25 sub-paths $(x_t)^i_{0 \le t \le T_j}$ where $j = 1, 2, \ldots, 25$ and $2 = T_1 < T_2 < \cdots < T_{25} = 1000$ are evenly spaced. Then we compute the MLE for each subpath, to see whether $\hat{\theta}_T \to \theta$ as $T \to \infty$. We will follow this procedure (slightly altered) several times in this report.

To see the asymptotic distribution of θ we simply sample 1000 independent discretized paths as before and compute the MLE on each from which we can compute and visualize its distribution.

From Figure 8 we clearly see that as $T \to \infty$ the MLE is approaching $\theta = 4$, and from Figure 9 we see the empirical distribution of the (rescaled) MLE are approaching a standard Gaussian distribution.

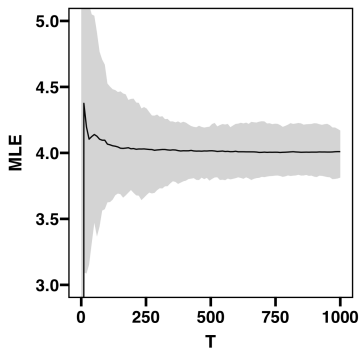

Figure 8: Mean and 95% inter-quartile range of MLE as T grows.

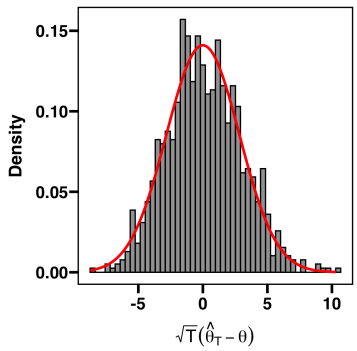

Figure 9: Histogram of MLE of 1000 independent samples from $(X_t)_{0 \le t \le 500}$. The red curve is the density of a $N(0, 2\theta)$ distribution.

4.1 MLE of drift matrix in d-dimensional OU process

Extending from one dimension to d dimensions is straightforward. Let for the remainder of the report $(X_t)_{0 \le t \le T}$ be a continuously observed path of the solution to $dX_t = -AX_t \, dt + dW_t$ with $A \in \mathbb{R}^{d \times d}$ and $(W_t)_{t \ge 0}$ a d-dimensional Brownian motion. Definition 4.2 is still the definition for a MLE. And we can, using Girsanov's theorem, compute the negative log-likelihood [2],

$$\ell_T(A) = -\log\left(\frac{dP_A^T}{dP_W^T}\right) = \int_0^T (AX_t)^\intercal \, dX_t + \frac{1}{2}\int_0^T (AX_t)^\intercal (AX_t) \, dt$$

Using that for two vectors $u, v \in \mathbb{R}^k$ for any $k \in \mathbb{N}$ we have $u^\intercal v = \mathrm{Tr}(vu^t)$ and that the trace essentially is just a sum so it "commutes" with the integral and is linear, we can rewrite ℓ_T

$$\ell_T(A) = \mathrm{Tr}\left(\int_0^T dX_t X_t^\intercal A^\intercal + \frac{1}{2} A \int_0^T X_t X_t^\intercal \, dt A^\intercal\right)$$

We name the integrals here as they will appear many times. For the sake of consistency with later sections we want to divide ℓ_T by T, so we let

$$\eta_T = \frac{1}{T}\int_0^T dX_t X_t^\intercal, \quad \hat{C}_T = \frac{1}{T}\int_0^T X_t X_t^\intercal \, dt. \tag{8}$$

And then we have,

$$\mathscr{L}_T(A) = \frac{1}{T}\ell_T(A) = \mathrm{Tr}\left(\eta_T A^\intercal + \frac{1}{2}A\hat{C}_T A^\intercal\right), \tag{9}$$

$$\nabla\mathscr{L}_T(A) = \eta_T + A\hat{C}_T \tag{10}$$

It follows that $\{\eta_T, \hat{C}_T\}$ constitutes a sufficient statistic for the model.

As our goal is to find the A minimizing the negative log-likelihood (maximizing the likelihood) we can equivalently minimize ℓ_T or $\mathscr{L}_T$. Solving the score equation $\nabla\mathscr{L}_T(A) = 0$ we get the maximum likelihood estimator (if $\hat{C}_T$ is invertible - which it is almost surely ([6], Theorem 4.1))

$$\hat{A}_{MLE} = -\eta_T\hat{C}_T^{-1} = -\int_0^T dX_t X_t^\intercal \left(\int_0^T X_t X_t^\intercal\, dt\right)^{-1}. \tag{11}$$

Again under (H1) a number of asymptotic results holds true. In particular we have that $\hat{A}_{MLE}$ is consistent $\hat{A}_{MLE} \xrightarrow{P} A$ and asymptotically normal $\sqrt{T}(\mathrm{vec}(\hat{A}_{MLE}) - \mathrm{vec}(A)) \xrightarrow{\mathscr{D}} N(0, C_\infty^{-1} \otimes I_{d^2})$. Note also that $\hat{C}_T \xrightarrow{a.s.} C_\infty$.

As done for the 1-dimensional MLE we also make a simulation study to see that the d-dimensional MLE is actually consistent. We follow the same procedure by taking independent samples from the OU process with a given drift matrix A and sub-paths hereof on which we compute the MLE (see Section 4.0.1).

We see from Figure 10 that as T grows, the maximum likelihood estimator becomes better and its distance in Frobenius norm from the true drift matrix seems to converge to 0 as $T \to \infty$ in support with our claim that the estimator is consistent.

We have seen above that MLE is consistent and asymptotically normal, but these properties are under an asymptotic regime and as we will show, the length of necessary observation T such that these properties are useful in practice, is highly dependent on the dimension of the process d. This is analogous to the fact that MLE in classical statistical models are best when the number of predictors are much lower than the number of observations.

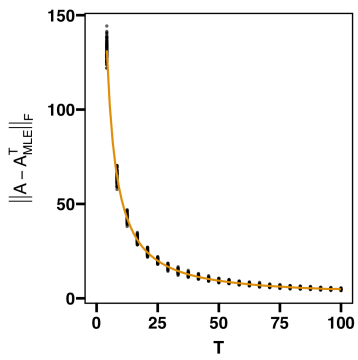

Figure 10: Difference in Frobenius norm between MLE and the true matrix for $T \to \infty$.

4.2 Curse of dimensionality

To see the breakdown of MLE for high dimensional processes observed over a limited time, we simulate drift matrices with dimension 5 to 50, for each of these we simulate 10 sample paths over $[0, T] = [0, 10]$ and compute the MLE for each. The Frobenius norm of the difference between the MLE and the true A, $\|A^d - A^d_{\mathrm{MLE}}\|_F$, are plotted in Figure 11.

We clearly see from Figure 11 that MLE is quite precise for low dimensional processes but the precision decays rather quickly for growing d. Furthermore, notice that for high values of d many MLEs are not even computed, which is due to $\hat{C}_T$ not being invertible. Although, as noted, $\hat{C}_T$ is a.s. invertible the problem probably is that the underlying discretization of $(X_t)_{0\le t\le 10}$ is not fine enough.

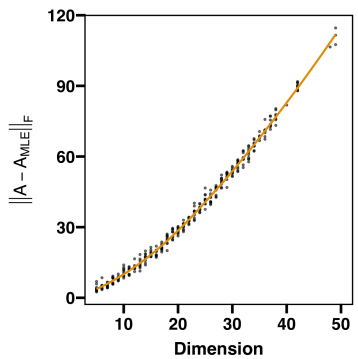

Figure 11: Difference in Frobenius norm between MLE and the true drift matrix for $d \to \infty$. The orange line is a smoothed mean curve.

4.3 Computation

The maximum likelihood estimator for the drift matrix in a d-dimensional Ornstein-Uhlenbeck process is implemented in two ways. The obvious

implementation is using the exact solution to the score-equation (11) which, besides computation of the integrals, requires inverting a $d \times d$ matrix and a matrix multiplication. This is quite efficient even for large d. Otherwise we could try to numerically optimize the log-likelihood (9). And since we have the gradient of the likelihood this can be done with traditional Newton or Quasi-Newton methods. We apply an algorithm called *L-BFGS* implemented in the `lbfgs R` package [8] as a minor modification of this algorithm will also handle optimization of several other presented estimators, for more details see Section 5.2.1. This is less efficient than the exact formula since we have to evaluate the likelihood and its gradient many times.

Both methods require computation of the sufficient statistics η_T and $\hat{C}_T$ (8). This will become a theme throughout the report, these statistics have to be computed many times. Hence for procedures involving several evaluations of the likelihood or its gradient (as the optimization described in the last paragraph) it is helpful to compute η_T and $\hat{C}_T$ beforehand which is possible since these are constants given the observed data. Hence in all implemented estimators for this project one can either supply them with T and (the discretized) $(X_t)_{0 \leq t \leq T}$ or with the precomputed η_T and $\hat{C}_T$.

5 The LASSO estimator

We have seen that working with maximum likelihood estimation for Ornstein-Uhlenbeck processes causes problems when $d \to \infty$ and T is fixed. But if we are willing to make further assumptions on our drift matrix A, high-dimensional inference might be feasible.

Returning to one of the applications of the d-dimensional Ornstein Uhlenbeck process, namely to model linear interactions in a network of actors, we noted that such networks would usually be sparse, as many of the actors would not interact at all. Thus to model this scenario we could restrict A to be sparse, and exploit this fact to create more suitable estimators.

The basic idea behind the LASSO estimator is to penalize the loss function with the ℓ_1-norm of the parameter. It is well known that penalizing with an ℓ_q "norm" where $q \leq 1$ (actually a norm when $q = 1$) induces sparsity in the parameter, but for $q < 1$ the ℓ_q norm is not convex and hence optimization is not computationally efficient with usual methods. So penalizing with the ℓ_1 norm is the best we can do in terms of sparsity if we still want a convex problem [9].

We still consider the continuously observed d-dimensional Ornstein-Uhlenbeck process (Definition 3.3), but for the remainder of the report we will assume that our drift matrix A is sparse,

(H3) $\|A\|_0 = \sum_{i,j \leq d} 1_{\{A_{ij} \neq 0\}} \leq \rho \cdot d^2 = s_0$ for some $s_0 \geq d$ (i.e. $\rho \in [\frac{1}{d}, 1]$).

We must have that $s_0 \geq d$ as (H1) otherwise would be violated. Let us now define the LASSO estimator,

Definition 5.1 *Let $\lambda > 0$, then the LASSO estimator is*

$$\hat{A}^L_\lambda = \underset{A \in \mathbb{R}^{d\times d}}{\operatorname{argmin}} \{\mathscr{L}_T(A) + \lambda \|A\|_1\} \tag{12}$$

As both $\mathscr{L}_T$ (due to $\hat{C}_T$ being a.s. positive definite) and $\|\cdot\|_1$ are strictly convex functions (a.s.), so is their sum which ensures that $\hat{A}^L_\lambda$ is (a.s.) unique [1]. Note that we have actually defined a class of estimators $\{\hat{A}^L_\lambda : \lambda > 0\}$ and we thus need to find the optimal λ. Although theoretical results for the choice of λ exists, these are in practice too conservative and is mainly used for theoretical analysis of the estimator. How to choose λ is considered in Section 7. The effect of λ is seen in Figure 12 where the LASSO estimator is computed for a sequence of λ values. We see that for increasing λ the sparsity of the solution increases until we arrive at the 0-matrix.

Note that the ℓ_1 norm is not differentiable at 0, hence we cannot simply find the derivative and solve the score equation. Instead we say that a matrix $G \in \mathbb{R}^{d\times d}$ is a subgradient of a convex function $f : \mathbb{R}^{d\times d} \to \mathbb{R}$ at the point $A_0 \in \mathbb{R}^{d\times d}$ if $f(A) - f(A_0) \geq \operatorname{Tr}(B^\intercal(A - B_0))$ for all $A \in \mathbb{R}^{d\times d}$. The set of all subgradients at a point A_0 is called the subdifferential at A_0, denoted $\partial f(A_0)$. Subgradients of the ℓ_1-norm, $G \in \partial||A_0||_1$ for some A_0 in particular satisfies $||G||_\infty \leq 1$. Furthermore, as $\mathscr{L}_T$ is differentiable with gradient (10), the subdifferential of $Q_\lambda(A) = \mathscr{L}_T(A) + \lambda \|A\|_1$ is $\partial Q_\lambda(A_0) = \nabla L_T(A_0) + \lambda\partial||A_0||_1$.

Now, a necessary and sufficient condition for a minimizer (12) is that $0 \in \partial Q_\lambda(\hat{A}^L_\lambda)$ where $Q_\lambda(\hat{A}^L_\lambda) = \mathscr{L}_T(\hat{A}^L_\lambda) + \lambda||\hat{A}^L_\lambda||_1$ is the LASSO loss function (called Karush-Kuhn-Tucker (KKT) condition [9]), which in particular implies that a LASSO solution $\hat{A}^L_\lambda$ satisfies

$$\left\|\eta_T + \hat{A}^L_\lambda \hat{C}_T\right\|_\infty \leq \lambda \tag{13}$$

[2]. We will discuss this property further in Section 6.

Under (H1-3), a number of oracle inequalities and error bounds on several norm-differences of the estimator and the true matrix have been shown, which under specific values of λ gives the asymptotic and non-asymptotic properties of the LASSO estimator. Such results are considered in [1, 2], we will only consider them briefly.

[1] proves that the LASSO estimator under consideration has asymptotically optimal estimation rate for a specific λ in line with the LASSO estimator in linear Gaussian models. Besides optimal estimation rates one would prefer if an estimator of a sparse model were consistent for variable selection, that is for an estimator $\hat{A}_T$

$$\mathbb{P}(\operatorname{supp}(\hat{A}_T) = \operatorname{supp}(A)) \to 1 \quad \text{as } T \to \infty \tag{14}$$

where $\operatorname{supp}(A) = \{(i, j) : A_{ij} \neq 0\}$. If an estimator is asymptotically both of optimal estimation rate and are consistent for variable selection we say it has the *oracle property* [10]. In linear Gaussian models it is well-known that for given λ the LASSO estimator cannot have both properties [11]. Whether this is also the case for sparse estimation of the drift matrix in Ornstein-Uhlenbeck processes has not been asserted to the knowledge of the author.

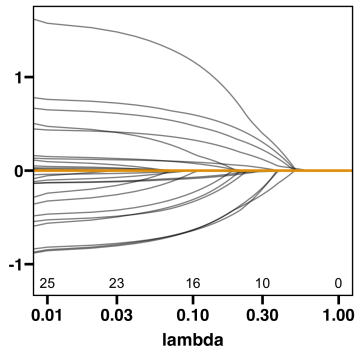

Figure 12: Solutionpath for the LASSO estimator of each entry in an $A \in \mathbb{R}^{5\times 5}$ sparse drift matrix for increasing lambda. The small number above the x-axis is the number of non-zero elements in the estimate. The left hand side corresponds to the MLE, the right hand side to the 0-matrix "estimator". Notice that the x-axis is on $\log_{10}$ scale.

5.1 Variations on LASSO

5.1.1 Different penalization norms

A natural extension of the LASSO is to use different penalizations. For example we could penalize by some general ℓ_p-norm to get the so called *Bridge estimator* proposed for linear models in [12]. Note that this has both the LASSO and Ridge estimators as special cases. But for $p > 1$ the ℓ_p-norm will not generally induce sparse solutions, and we would thus need some thresholding procedure to determine the sparse components. At the same time the ℓ_p norm for $p > 1$ will penalize parameters of large magnitude more than those of small magnitude, which we will see below is not a desirable property in our case. See below for considerations on $p < 1$. We will not consider the Bridge estimator any further, for more details see [13].

Likewise we could try to combine different ℓ_p-norms in a single penalization. Combining the ℓ_1 and ℓ_2 norm is proposed for linear models as the *Elastic-net estimator* in [14]. There are several good reasons for using Elastic-net in linear models (for example when having highly correlated predictors or many more predictors than observations), but none of these directly translates into a good reason for using it in sparse estimation of the drift matrix in an Ornstein-Uhlenbeck process. However, the estimator is very simple to implement so we will see how it compares to the LASSO. We thus define the Elastic-Net estimator for estimation of the drift matrix in an OU process,

Definition 5.2 *Let $\lambda > 0$ and $\alpha \in [0, 1]$, the Elastic-net estimator is,*

$$\hat{A}^{EN}_{\lambda,\alpha} = \underset{A \in \mathbb{R}^{d\times d}}{\operatorname{argmin}} \left\{ \mathscr{L}_T(A) + \lambda \left((1-\alpha)\frac{\|A\|_2^2}{2} + \alpha \|A\|_1 \right) \right\}. \tag{15}$$

Note that we can rewrite this loss into a differentiable function plus an ℓ_1 penalization,

$$\hat{A}^{EN}_{\lambda,\alpha} = \underset{A \in \mathbb{R}^{d\times d}}{\operatorname{argmin}} \left\{ \underbrace{\mathscr{L}_T(A) + \lambda(1-\alpha)\frac{\|A\|_2^2}{2}}_{\text{Differentiable}} + \lambda\alpha \|A\|_1 \right\} \tag{16}$$

and in particular the gradient of the differentiable part becomes $\eta_T + A\hat{C}_T + \lambda(1-\alpha)A$.

Note that this estimator has both the LASSO and Ridge estimators as special cases, we can in particular view Elastic-net as a compromise between the ℓ_1 and ℓ_2 penalization where the tuning-parameter α controls the weighting of each.

5.1.2 Two-step estimators

LASSO is an acronym for **L**east **A**bsolute **S**hrinkage and **S**election **O**perator. We have discussed the selection part, namely that some parameters are estimated to 0 and thus not "selected". Meanwhile, LASSO is also a shrinkage estimator, meaning that even the parameters which are not estimated to 0 are biased toward 0 and in particular they are not in general consistent. This is mainly due to all of the parameters being equally penalized when penalizing with the ℓ_1-norm, which is a problem that theoretically could be solved as noted by penalizing with an ℓ_p-norm for $p < 1$, but this leads to non-convex loss functions which are

in particularly hard to optimize in high dimensions. [11, 15]. We consider two possible work-arounds to reduce bias which continues to only work with convex loss functions.

[15] suggests for linear models a two-step procedure called *relaxed LASSO* which identifies the support set of the parameter using LASSO, and then applying LASSO once again, but only considering the parameters in the support set. They also showed that the relaxed LASSO has a faster estimation rate than LASSO for optimal choice of tuning parameters and leads to sparser estimates than LASSO.

We propose a relaxed LASSO estimator for estimating the sparse drift matrix A from a continuously observed Ornstein-Uhlenbeck process $(X_t)_{0\leq t\leq T}$,

Definition 5.3 *Let $\lambda > 0$ and $\phi \in [0,1]$, the relaxed LASSO estimator is*

$$\hat{A}^{RL}_{\lambda,\phi} = \underset{A\in\mathcal{M}_\lambda}{\operatorname{argmin}}\{\mathcal{L}_T(A) + \phi\lambda \|A\|_1\} \tag{17}$$

where $\mathcal{M}_\lambda = \{A \in \mathbb{R}^{d\times d} : \operatorname{supp}(A) = \operatorname{supp}(\hat{A}^L_\lambda)\}$ where $\hat{A}^L_\lambda$ is an initial LASSO estimate.

Note that this estimator is identical to LASSO if $\phi = 1$. The case where $\phi = 0$ corresponds to the non-zero elements of matrices in $\mathcal{M}_\lambda$ are estimated using maximum likelihood estimation. The loss function is still strictly convex and thus have unique minimum given λ and ϕ. The parameters λ and ϕ are to be chosen for example using cross-validation, see Section 7 for more details. To the knowledge of the author no results on this estimator exists for estimation in Ornstein-Uhlenbeck processes.

[11] suggests for linear models a reweighing of the parameters in the penalization using an initial estimator called *adaptive LASSO* such that those whose initial estimate are of large magnitude are penalized less than those of small magnitude. They showed that the adaptive LASSO has the oracle properties. For Ornstein-Uhlenbeck processes [1] proposed an adaptive LASSO estimator of the drift matrix reweighing by the MLE,

Definition 5.4 *Let $\lambda > 0$ and $\gamma \geq 0$, the adaptive LASSO estimator is,*

$$\hat{A}^{AL}_{\lambda,\gamma} = \underset{A\in\mathbb{R}^{d\times d}}{\operatorname{argmin}} \left\{\mathcal{L}_T(A) + \lambda \left\|A \circ |\hat{A}_{MLE}|^{\circ(-\gamma)}\right\|_1\right\} \tag{18}$$

where the product $(A,B) \mapsto A \circ B$, absolute value $|\cdot|$ and exponentiation $(A,\gamma) \mapsto A^{\circ\gamma}$ are all elementwise (Hadamard-type operations).

The entries in the MLE solution is almost surely non-zero, so they can be inverted almost surely and no trouble arises from the definition. Given the MLE, λ and γ the problem is still strictly convex and has unique solution. It is further shown in [1] (Theorem 4) that $\hat{A}^{AL}_{\lambda,\gamma}$ has the oracle property when restricted to the support set $\operatorname{supp}(A)$ of the true A.

5.2 Computation

5.2.1 Optimization procedure

The main optimization in each of the LASSO variants considered above are convex and hence relatively efficient. But as noted, the loss functions are not differentiable at all points and hence traditional gradient-descent-like methods involving iterative computation of the gradient is not possible. Many methods have been developed which are quite efficient

for the minimization of these ℓ_1 penalized loss functions, these include proximal methods [16], coordinate descent methods [17] and least-angle regression [18].

We have used an algorithm called *Orthant-Wise Limited-memory Quasi-Newton (OWL-QN)* [19] which are specifically developed for minimizing loss functions of the form

$$f(x) = \ell(x) + \lambda \|x\|_1 . \tag{19}$$

under the assumptions that ℓ is convex, bounded from below and that $\nabla\ell$ is L-Lipschitz for some $L > 0$. Our negative log-likelihood function is clearly convex and bounded from below (it takes its minimum at the MLE). Furthermore $\nabla\mathscr{L}_T$ are affine and thus in particular Lipschitz.

Essentially this method exploits that in each orthant (which is the general n-dimensional analogue to the Euclidean quadrant) the ℓ_1-norm is actually differentiable since no elements of the parameter changes sign when staying inside the orthant, furthermore it is linear and will thus not affect the second-order behaviour of the loss. Thus given an initial point, find its orthant and use a quasi-Newton step restricting the next point to have the same signs as the initial one, after the quasi-Newton we can then allow for the orthant to change. The quasi-Newton method could for example be *L-BFGS*. For more details see [19].

This method is quite efficient even for parameter spaces of very high dimensions and has an easy to use `R` implementation in the package `lbfgs` [8]. It should be noted that the proofs of convergence results in [19] have been found to be flawed [20]. A modified version of the algorithm, and a rigoruos convergence analysis of it, was propsed and made in [20] but is not implemented in the `lbfgs` package. Meanwhile several studies have still shown the original OWL-QN algorithm to work quite well [20]. In particular we have found the OWL-QN to be much faster than the alternative derivative-free methods such as Nelder-Mead which does not utilize the information in $\nabla\mathscr{L}_T$.

5.2.2 Implementing estimators

Implementing the LASSO estimator using OWL-QN and the `lbfgs` package is straightforward as the LASSO loss (12) is exactly on the required form (19) after necessary vectorization of the parameter matrix to fit into the implementation. The optimization is initialized in the maximum likelihood estimator and ran with the default `lbfgs` convergence parameters until convergence is reached. The Elastic-net estimator (15) is likewise easy to implement as the loss function is easily rewritten (16) to the desired form (19). Implementing the two step LASSO estimators with OWL-QN requires a bit more consideration as they do not fit directly into (19).

The relaxed LASSO starts by fitting a LASSO just as described above with tuning parameter $\lambda > 0$. Then we fit another LASSO estimator but only on the non-zero elements of the initial estimator now with a scaled tuning parameter $\phi\lambda$ where $\phi \in [0, 1]$ and λ as before. In practice we need to optimize over a vector of dimension $\rho \cdot d^2$ where ρ is the density of non-zero elements of $\hat{A}^L_\lambda$, which is then embedded into a $d \times d$ matrix at the right positions for the computation of the likelihood.

For the adaptive LASSO we start by computing the MLE. Then for the loss to fit into the OWL-QN algorithm we start by observing that

$A = A \circ |\hat{A}_{MLE}|^{\circ(-\gamma)} \circ |\hat{A}_{MLE}|^{\circ\gamma}$ for all $A \in \mathbb{R}^{d\times d}$ and we can thus rewrite our loss function (18) to

$$\mathscr{L}_T(A \circ |\hat{A}_{MLE}|^{\circ(-\gamma)} \circ |\hat{A}_{MLE}|^{\circ\gamma}) + \lambda \left\| A \circ |\hat{A}_{MLE}|^{\circ(-\gamma)} \right\|_1$$

Thus we could start by solving a different optimization, namely if we have computed the MLE and fixed $\gamma > 0$, we have the optimization, which is on the desired form (19),

$$\hat{A}^{\mathrm{AL}*}_{\lambda,\gamma} = \underset{A\in\mathbb{R}^{d\times d}}{\operatorname{argmin}} \left\{ \mathscr{L}_T(A \circ |\hat{A}_{MLE}|^{\circ\gamma}) + \lambda \|A\|_1 \right\}$$

which corresponds to estimating $A \circ |\hat{A}_{MLE}|^{\circ(-\gamma)}$ by $\hat{A}^{\mathrm{AL}*}_{\lambda,\gamma}$. But as $|\hat{A}_{MLE}|^{\circ(-\gamma)}$ is completely known we can compute the adaptive LASSO estimator as,

$$\hat{A}^{\mathrm{AL}}_{\lambda,\gamma} = \hat{A}^{\mathrm{AL}*}_{\lambda,\gamma} \circ |\hat{A}_{MLE}|^{\circ\gamma}.$$

6 Dantzig estimator

We saw in the discussion leading up to and concluding with (13) that the LASSO estimator $\hat{A}^L_\lambda$ satisfies

$$\left\| \eta_T + \hat{A}^L_\lambda \hat{C}_T \right\|_\infty \leq \lambda$$

Now wanting to get the most sparse possible estimate (in terms of ℓ_1 norm) we could assume that this condition is actually satisfied and under this constraint minimize the ℓ_1 norm of the parameter. This idea leads to the Dantzig estimator as proposed in [2],

Definition 6.1 *Let $\lambda > 0$, then the Dantzig estimator is*

$$\hat{A}^D_\lambda = \underset{A\in\mathbb{R}^{d\times d}}{\operatorname{argmin}} \left\{ \|A\|_1 : \left\| \eta_T + A\hat{C}_T \right\|_\infty \leq \lambda \right\} \tag{20}$$

The Dantzig estimator is closely related to the LASSO estimator, and in particular given λ we have that $||\hat{A}^D_\lambda||_1 \leq ||\hat{A}^L_\lambda||_1$ since $\hat{A}^L_\lambda$ satisfies the constraint and is thus a possible solution to (20) although not necessarily the optimal. Exactly as the LASSO the higher λ is the more sparse the corresponding estimate, see Figure 13.

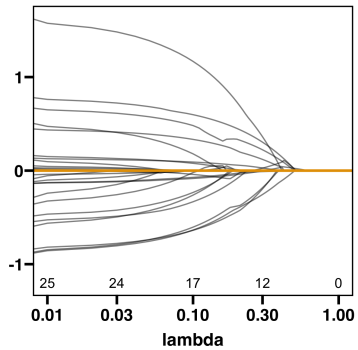

Figure 13: Solutionpath for the Dantzig estimator of each entry in an $A \in \mathbb{R}^{5\times5}$ sparse drift matrix for increasing lambda. The small number above the x-axis is the number of non-zero elements in the estimate. The left hand side corresponds to the MLE, the right hand side to the 0-matrix "estimator". Notice that the x-axis is on $\log_{10}$ scale.

Like for the LASSO, a number of oracle inequalities and error bounds on ℓ_1 and ℓ_2 norm-difference between the estimator and the true matrix have been found for the Dantzig estimator of the sparse drift matrix in Ornstein-Uhlenbeck processes. In particular these are proven and used to show that the LASSO and Dantzig estimators have equivalent performances in [2], which is also in line with them being asymptotically equivalent in linear models [21].

6.1 Computation

The Dantzig estimator can be implemented as a constrained linear optimization problem using linear programming. But before this is apparent we have to reformulate the problem somewhat. First of all, the objective $A \mapsto ||A||_1$ is not linear, but note that we can write

$||a||_1 = \sum_{i=1}^{k} |a_i| = \sum_{i=1}^{k} a_i^+ + a_i^-$ for any $a \in \mathbb{R}^k$, $k \in \mathbb{N}$ where $a_i^+ = \max\{a_i, 0\}$, $a_i^- = \max\{-a_i, 0\}$ for all $i \in \{1, ..., k\}$ are the vectors of positive respectively negative parts of a. This also means that if we vectorize our parameter matrix A we can write

$$\|A\|_1 = 1_{2\cdot d^2}^\intercal \begin{pmatrix} \text{vec}(A)^+ \\ \text{vec}(A)^- \end{pmatrix} = c^\intercal y \quad \text{where } c^\intercal = 1_{2\cdot d^2}^\intercal = (1 \;\; 1 \;\; \cdots \;\; 1) \in \mathbb{R}^{2\cdot d^2}$$

which is now a linear objective in y, and we can easily identify A from $(\text{vec}(A)^+ \;\; \text{vec}(A)^-)^\intercal$ as $\text{vec}(A) = \text{vec}(A)^+ - \text{vec}(A)^-$ and de-vectorize.

Likewise we can rewrite the constraint into a linear inequality constraint. We start by vectorizing

$$||\eta_T + A\hat{C}_T||_\infty = ||\text{vec}(\eta_T) + \text{vec}(A\hat{C}_T)||_\infty \leq \lambda$$

We can now write $\text{vec}(A\hat{C}_T) = (\hat{C}_T^\intercal \otimes I_d)\,\text{vec}(A) = (\hat{C}_T \otimes I_d)\,\text{vec}(A)$ so let $K = \hat{C}_T \otimes I_d$. And as the ∞-norm is just the maximum absolute value we must equivalently have that every entry's absolute value is less than λ, or

$$-\lambda 1 \leq \text{vec}(\eta_T) + K\text{vec}(A) \leq \lambda 1_{d^2}$$

Rewriting these inequalities and splitting $\text{vec}(A)$ up into the difference of its positive and negative parts again gives us,

$$\begin{pmatrix} -K\,(\text{vec}(A)^+ - \text{vec}(A)^-) \\ K\,(\text{vec}(A)^+ - \text{vec}(A)^-) \end{pmatrix} \leq \begin{pmatrix} \lambda 1_{d^2} + \text{vec}(\eta_T) \\ \lambda 1_{d^2} - \text{vec}(\eta_T) \end{pmatrix}$$

which finally leads to a linear inequality constraint,

$$\underbrace{\begin{pmatrix} -K & K \\ K & -K \end{pmatrix}}_{=:M} \underbrace{\begin{pmatrix} \text{vec}(A)^+ \\ \text{vec}(A)^- \end{pmatrix}}_{=:y} \leq \underbrace{\begin{pmatrix} \lambda 1_{d^2} + \text{vec}(\eta_T) \\ \lambda 1_{d^2} - \text{vec}(\eta_T) \end{pmatrix}}_{=:b} \tag{21}$$

Thus we see that our original optimization (20) can be written as a linear program on standard form, using the names defined in (21)

$$\text{minimize } c^\intercal y \quad \text{subject to: } My \leq b$$

where $c = 1_{2\cdot d^2}$. Solving linear programs are computationally very efficient and implementing the estimator is simple when using for example the `R` package `lpSolve` [22] using the simplex method [23].

7 Choosing tuning parameters

All of the presented estimators, with MLE being the exception, in fact define a class of estimators parametrized by their tuning parameters λ, α, γ and ϕ. For the LASSO and Dantzig estimators, theoretically obtained values of λ are suggested in [1, 2] but it is well known that tuning parameters from such theoretical considerations are often very pessimistic [9]. The classical way of choosing tuning parameters are instead to do cross-validation [24], which divides the data, often several times, into training data on which you fit a family of models and validation data on which you test the models using some loss function or prediction error and choosing the model which fits the validation data best. The

idea is that testing the models on the validation data prevents overfitting by choosing tuning-parameters from the data on which you have fitted the model.

Classical cross-validation procedures would however not work in our case as they assume discretely observed i.i.d. data, which is certainly not the case for our continuous observation of the Ornstein-Uhlenbeck process where we have to take into account the autocorrelation of the process. Instead [1] propose to do a simple holdout validation data strategy, where we compute the estimators for each combination of tuning-parameters on the first 80% of the observed process and validate these estimators on the last 20% using the log-likelihood loss as validation metric. [2] proposes a similar strategy using 90% of the observed process as training data and using the log-likelihood loss divided by the ℓ_1-norm of the estimate as validation metric. In our experience both strategies yield similar results, and no apparent reason for the more comlicated validation metric are given in [2] hence we use the the log-likelihood loss as validation metric.

That is, for optimal estimation of each presented estimator, fit for each combination of tuning parameters necessary for the given estimator,

$$\hat{A}^{\varepsilon}_{\tau} = \operatorname*{argmin}_{A \in \mathbb{R}^{d \times d}} Q^{\varepsilon}_{\tau}(A; 0.8 \cdot T)$$

where $Q^{\varepsilon}_{\tau}(A; 0.8 \cdot T)$ is the loss corresponding to model ε computed for the training data (first 80% of the whole observation) with tuning-parameters τ, where

$$\begin{aligned} \varepsilon \in \mathscr{E} = \{&\text{LASSO, Elastic-net, Adaptive LASSO,} \\ &\text{Relaxed LASSO, Dantzig}\} \\ \tau \in \mathscr{T}_{\varepsilon} = \{&\text{All possible combinations of tuning-parameters in } \varepsilon\}. \end{aligned}$$

Then we test each estimator, choosing the tuning-parameters minimizing the loss,

$$\hat{\tau} = \operatorname*{argmin}_{\tau \in \mathscr{T}_{\varepsilon}} \mathscr{L}_{(0.8 \cdot T, T]}(\hat{A}^{\varepsilon}_{\tau}).$$

from which we gain the cross-validated estimator $\hat{A}^{\varepsilon}_{\hat{\tau}} = \operatorname{argmin}_{A \in \mathbb{R}^{d \times d}} Q^{\varepsilon}_{\hat{\tau}}(A; T)$.

In practice an extensive global optimization of τ is not feasible, hence we search for the optimal τ on a fine grid in which we believe it lies. For out simulation studies to follow we use for each tuning parameter a grid with spacing of 0.01 and let the endpoints be apaptive for the tuningparameters under investigation. Furthermore for simplicity of the code, we translate the integrals computed in $\mathscr{L}_{(0.8 \cdot T, T]}$, such that for generic f, g (possibly stochastic processes)

$$\int_{0.8 \cdot T}^{T} f(t)\, dg(t) = \int_{0}^{0.2 \cdot T} f(t + 0.8 \cdot T)\, dg(t + 0.8 \cdot T).$$

8 Comparison of estimators

We have presented several different estimators, in particular the LASSO and Dantzig estimators, for the sparse drift matrix in an observed Ornstein-Uhlenbeck process and will now compare them through simulation studies.

Table 1: Norm errors in Frobenius and 1-norm. The first two columns are direct difference of the true and the estimated matrix. The middle two are the norm difference on the active set of the true matrix $\mathcal{A}$. The two last columns are the norm difference on the active set of the estimated matrix $\mathcal{AE}$.

	$\|\|A - \hat{A}^{\varepsilon}\|\|_F$	$\|\|A - \hat{A}^{\varepsilon}\|\|_1$	$\|\|A\|_{\mathcal{A}} - \hat{A}^{\varepsilon}\|_{\mathcal{A}}\|\|_F$	$\|\|A\|_{\mathcal{A}} - \hat{A}^{\varepsilon}\|_{\mathcal{A}}\|\|_1$	$\|\|A\|_{\mathcal{AE}} - \hat{A}^{\varepsilon}\|_{\mathcal{AE}}\|\|_F$	$\|\|A\|_{\mathcal{AE}} - \hat{A}^{\varepsilon}\|_{\mathcal{AE}}\|\|_1$
$\hat{A}_{MLE}$	7.57	86.88	5.22	32.88	7.57	86.88
$\hat{A}^{L}$	2.91	21.43	**2.73**	16.73	2.14	14.63
$\hat{A}^{EN}$	3.18	25.09	3.01	19.61	2.62	19.05
$\hat{A}^{AL}$	**2.83**	**19.63**	**2.72**	**16.41**	2.04	12.80
$\hat{A}^{RL}$	3.66	27.07	3.37	20.98	**1.85**	**10.75**
$\hat{A}^{D}$	3.93	31.72	3.60	23.49	3.05	21.12

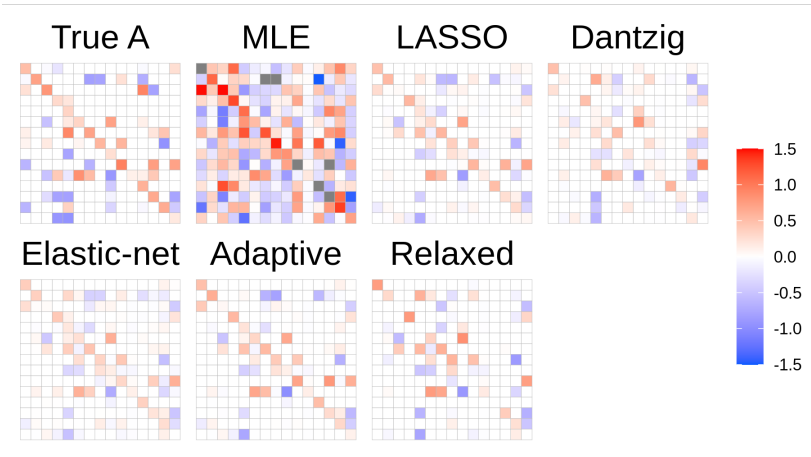

Figure 14: The true matrix and estimates using each method. Note that some element of the MLE are grey, these are all estimated higher than 1.5 which is the highest end on the colorscale, this makes the other estimates easier to see.

8.1 Performance on a given drift matrix

First and foremost we want to see how each estimator performs given a simulated process from a given sparse drift matrix. Hence we generate a sparse drift matrix $A \in \mathbb{R}^{15\times 15}$ with non-zero element density $\rho = 0.3$ and simulate a sample path over $[0, 20]$ from the corresponding Ornstein-Uhlenbeck process. We then perform cross-validation for each estimator giving us an estimate of the optimal tuning-parameters, which we use to fit each estimator to the full path. In Figure 14 each estimated matrix are shown using a heatmap to show the elementwise numerical differences and clearly see the sparsity. In Table 1 a number of error norms are presented for each estimate.

We see from Figure 14 that all of the penalized estimates have attained a high degree of sparsity close to that of the true matrix, meanwhile is the MLE as always non-sparse. The MLE is furthermore overestimating even the parameters in the active set of the true matrix, this is likely caused by the low sample size $T = 20$ as the MLE is consistent as $T \to \infty$. Meanwhile we also see the schrinkage in action, as in particular the LASSO and Elastic-net estimators seem to underestimate the elements in the active set. From Figure 14 we also see that the penalized estimates all have identified many of the same elements as sparse, but the some of the active elements have somewhat different values.

From Table 1 we see that all of the penalized methods perform quite well even with our small sample size, this is especially true in comparison to the MLE which in all different metrics shown here are considerably

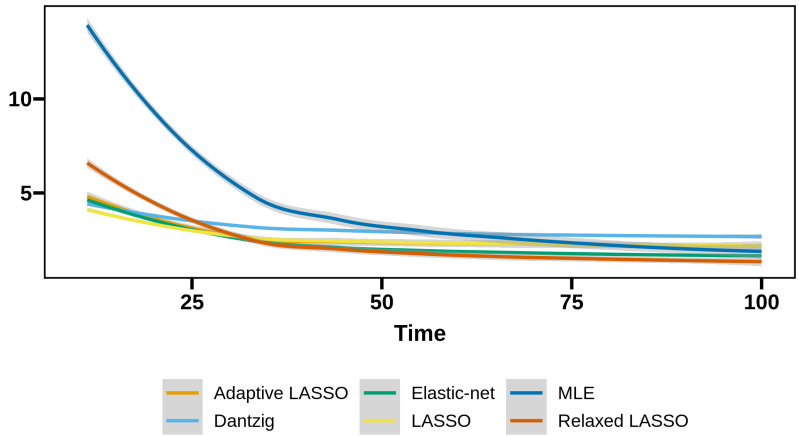

Figure 15: Distance in Frobenius norm ± standard deviation from each estimator to the true drift matrix.

worse. The adaptive LASSO seems to have a slight edge over the other estimators both when looking at the direct difference between estimates and the true matrix and when restricting the difference to be on the active set of the true matrix in ℓ_1 norm, but looking at the Frobenius norm, it is very comparable to the LASSO. Meanwhile when looking at the elements which were actually selected by the estimators then the relaxed LASSO performs better.

These results would of course have been more interesting had we sampled several different paths from the OU process and gained some knowledge about the uncertainty in the estimates, but due to limited computational power and time this was not feasible.

8.2 $T \to \infty$

We now investigate how the estimators perform asymptotically. For this purpose we generate a sparse drift matrix $A \in \mathbb{R}^{15\times 15}$ with $\rho = 0.3$ and follow the scheme outlined in Section 4.0.1 for each estimator. We sample 10 independent discretized paths over $[0, 100]$ with $\Delta t = 0.01$. Due to limited computational power and time, tuning-parameters are not chosen via cross-validation as they optimally should be at each time-point. Hence we have chosen rather arbitrary tuning-parameters but kept the ℓ_1 penalization parameter constant, in particular λ is constant for all but the Elastic-net estimator for which it is scaled together with α, for the results to be somewhat comparable.

We see in Figure 15 that for $T \in [0, 50]$ the penalized methods are performing noticeably better than the MLE whereafter they all seem to perform somewhat equivalently. While the Relaxed LASSO and Elastic-net seems to outperform both the MLE and the other estimators including the LASSO and Dantzig estimators which error seems to stabilize at some point > 0, this is likely a result of tuning-parameters being chosen constant and not adaptive with T. Overall every estimator gets better performance as T grows, but the gain in performance is noticeably higher for the MLE and somewhat for the relaxed LASSO than for the other penalized methods. The equivalence of the penalized methods is furthermore in accordance with the theoretical and numerical results of [1, 2] on the MLE, Dantzig, Adaptive LASSO and LASSO.

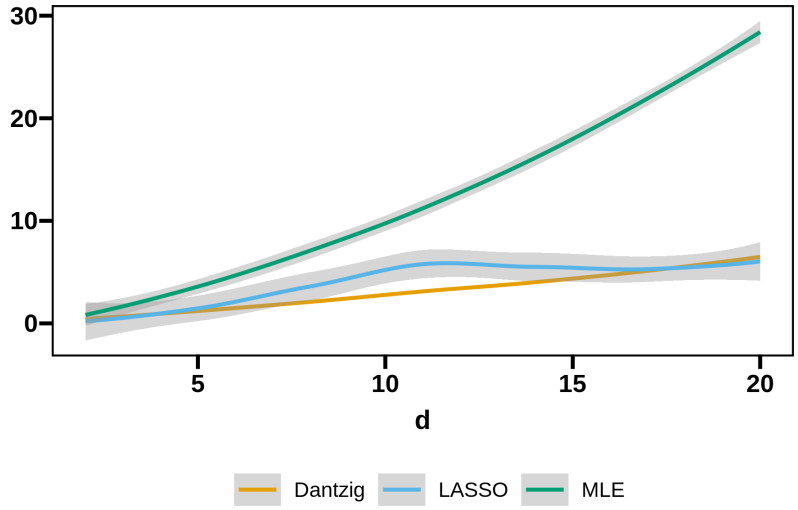

Figure 16: Frobenius norm difference ± standard deviation between estimated and true matrix for growing d

8.3 $d \to \infty$

We will also give a demonstration of the considerably better performance of our proposed estimators over the MLE as $d \to \infty$. Thus for $d = 2, 5, 8, \ldots, 20$ we sample matrices $A^d \in \mathbb{R}^{d \times d}$ with $\rho = 0.3$ and for each simulate 5 independent sample paths from their respective OU process. For each sample path we compute our estimates. Due to limited computational power and time, we have chosen only to show this for the LASSO and Dantzig estimators, which are the main focus of the report. Likewise we have chosen a constant λ for both estimators to avoid running multiple cross-validations.

We have already seen Section 4.2 that the performance of the MLE deteriorates as $d \to \infty$ and this is also seen in Figure 16 which also shows that this is much less of a concern with the LASSO and Dantzig estimators, for which the increase in error per extra dimension of the matrix is much lower than for the MLE.

8.4 Future work and limitations

We have seen comparisons between our presented estimators above, all of which have been done with a quite small number of replications and with a limited value of d and the discretization step size Δt. Thus more interesting and precise estimates might be found in the further simulation studies. Theoretical results e.g. on estimation rates and support recovery exist on some of our presented estimators but to the knowledge of the author not on the Relaxed LASSO or the Elastic-net estimators, which present possible extensions of the current theory.

Another interesting extension is looking at processes not satisfying (H1), which have been done for classical likelihood estimation for OU processes [25], but for which no work has been done in sparse high-dimensional estimation. There are also many other diffusion processes for which estimators such as LASSO and Dantzig could be defined and studied, but as noted in [2] the theoretical work on the high-dimensional estimators for OU processes highly relies on the linearity of the drift.

More efficient and user-friendly implementations of the sparse estimators presented here also pose possible future work. This might include extensions of commonly used LASSO optimization algorithms used for

linear models to the OU process models and other stochastic process models [16]. The generation of sparse matrices with positive eigenvalues are also open for more theoretically justified or more efficient methods.

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